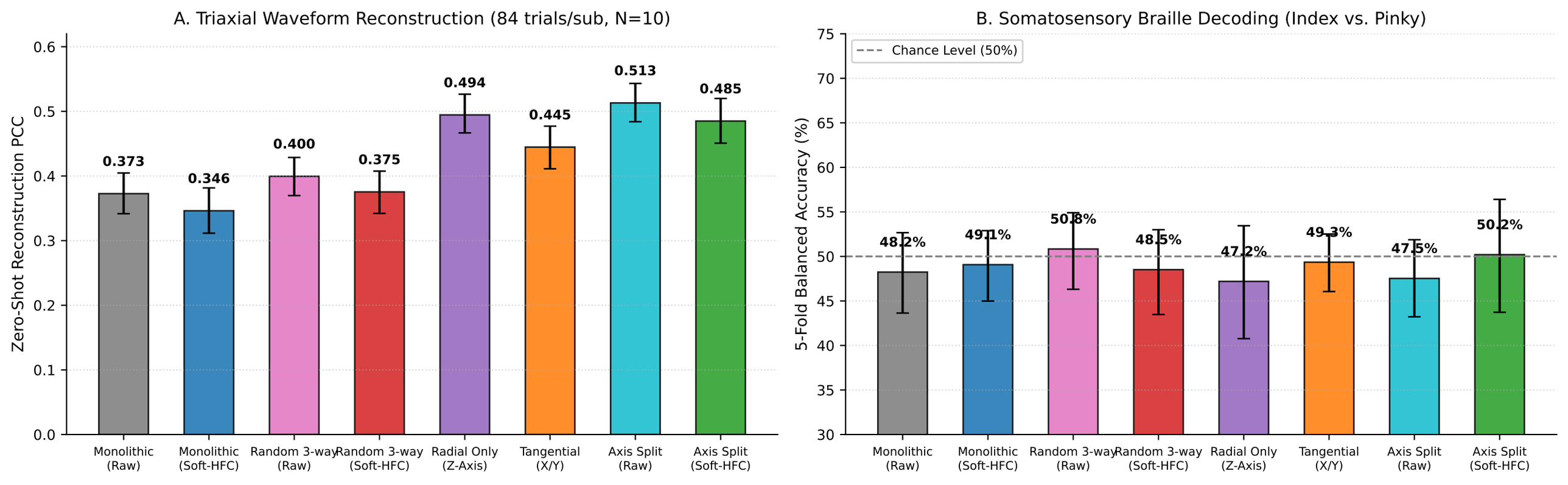
A. Triaxial Waveform Reconstruction (84 trials/sub, N=10)
Zero-Shot Reconstruction PCC
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.373
0.346
0.400
0.375
0.494
0.445
0.513
0.485
Monolithic (Raw)
Monolithic (Soft-HFC)
Random 3-way (Raw)
Random 3-way (Soft-HFC)
Radial Only (Z-Axis)
Tangential (X/Y)
Axis Split (Raw)
Axis Split (Soft-HFC)
B. Somatosensory Braille Decoding (Index vs. Pinky)
5-Fold Balanced Accuracy (%)
Chance Level (50%)
30
35
40
45
50
55
60
65
70
75
48.2%
49.1%
50.8%
48.5%
47.2%
49.3%
47.5%
50.2%
Monolithic (Raw)
Monolithic (Soft-HFC)
Random 3-way (Raw)
Random 3-way (Soft-HFC)
Radial Only (Z-Axis)
Tangential (X/Y)
Axis Split (Raw)
Axis Split (Soft-HFC)